Disease Prediction Using Ensemble Learning

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(Estd. u/s 3 of UGC Act 1956)



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OUTLINE

- SCOPE,OBJECTIVES, ABSTRACT, REQUIREMENT SPECIFICATION(HARDWARE AND
  - SOFTWARE & METHODS IDENTIFIED) - PROCESS MODEL,BLOCK DIAGRAM OF THE SYSTEM
- HTA, INTERACTION DESIGN(4 STEPS)STAKEHOLDERS IDENTIFICATION, STORYBOARDING,USE CASE MODELLING,PERSONAS,STATE TRANSITION NETWORK
- APPLY OF SCHENIDEMAN'S/NORMAN'S/NIELSEN'S, HERISTIC EVALUATION,KLM, GOMS
  - -GENERATE A TABLE AND GRAPH
- COMMUNICATION AND COLLABORATION, GROUPWARE
- VALIDATION-USABILITY TESTING, INTERFACE TESTING,USER ACCEPTANCE TESTING
  - USING ANY OPEN SOURCE TOOL
- RESULTS AND DISCUSSION-GRAPH GENERATION
- CONCLUSION
  - CONCLUDING WITH GIVING DETAILS OF IMPROVEMENT IN PERFORMANCE AND APPLICABILITY OF THE SYSTEM IN REAL TIME ENVIRONMENT
- Paper Template and related details

SCOPE

The scope of our project, centered around disease prediction through Ensemble Learning, extends across a diverse spectrum of health conditions. By integrating three robust algorithms – Decision Tree, Random Forest, and Gradient Boosting – through a unified Voting Classifier, we aim to enhance the accuracy and reliability of our predictive model.

Beyond the realm of early disease detection, our project holds the potential to revolutionize healthcare by providing valuable insights for personalized treatment plans. Leveraging artificial intelligence to analyze individual patient data enables us to predict the most effective treatments, contributing to superior patient outcomes and potentially reducing healthcare costs.

While emphasizing the transformative impact on disease diagnosis and treatment, we are committed to upholding ethical standards and implementing stringent data privacy measures.

Objectives

Our primary goal is to develop an intelligent diagnostic system capable of predicting diseases based on user-input symptoms. By doing so, we aim to alleviate the burden on hospital OPDs, reducing the workload on medical staff and providing users with a convenient alternative for preliminary health assessments. To achieve this, we plan to employ three distinct algorithms to enhance the reliability and accuracy of our predictive model.

In addition to its predictive capabilities, our system will feature a seamless and user-friendly interface, ensuring ease of use for individuals seeking quick health insights. We recognize the importance of data collection for continuous improvement. Hence, our system will include a database to store user-entered information, facilitating the development of future iterations.

Beyond functionality, our project places a strong emphasis on user comprehension. We will implement various visual representations of both collected data and model results, fostering a clearer understanding of the diagnostic process. By integrating these features, we aspire to create a sophisticated yet accessible healthcare tool that not only serves its immediate purpose but also lays the groundwork for ongoing enhancements and advancements in the field of predictive healthcare technology.

Abstract

In response to evolving health information needs globally, our project employs a combination of advanced machine learning algorithms—Decision Tree, Random Forest, and Gradient Boosting—to predict diseases accurately based on user-input symptoms. Targeting an accuracy range of 92-95%, this predictive system has the potential to revolutionize healthcare by enabling early disease prediction.

By facilitating proactive preventive measures, the model aligns with the adage "Prevention is better than cure." Our supervised machine learning approach empowers individuals to assess their health, make informed decisions, and reduce unnecessary pressure on hospitals. The system's efficiency lies in its ability to predict diseases, offering a valuable tool for users while contributing to smarter healthcare solutions.

The final model used is the Voting Classifier that combines 3 separate models. It exhibits an accuracy of 95%.

The User Interface (UI) of our system has been meticulously designed to provide a seamless and user-friendly experience to individuals seeking health insights. The interface is intuitively crafted, ensuring that users can effortlessly navigate through the system. The design prioritizes clarity and simplicity, allowing users to input their symptoms easily and interpret the results without any complexity.

Requirement Specification

Requirement Analysis for Developing an AI-Based Disease Prediction System in Healthcare

Requirement analysis is a crucial initial step in the development of an AI-based disease prediction system for healthcare. This process involves the identification and definition of the system's requirements and constraints, encompassing data sources, algorithms, and user interfaces. Below are key requirements for the effective development of such a healthcare AI system:

1. Data Sources:
   - Identify diverse and representative data sources such as electronic health records, medical imaging data, laboratory test results, genomics data, and patient-generated health data.
   - Ensure the selected data is comprehensive to train the AI model accurately and effectively.

2. Algorithm Selection:
   - Choose the appropriate algorithm for disease prediction based on the available data and the specific disease being targeted.
   - Options include logistic regression, decision trees, random forests, and neural networks.

3. Model Training and Validation:
   - Train the AI model using a large dataset with labeled data.
   - Validate the model's performance using a separate dataset to ensure accuracy and generalizability.

4. User Interface:
   - Design a user-friendly, intuitive interface accessible to both healthcare professionals and patients.
   - Provide clear and concise information regarding predicted disease risk and recommended interventions.

5. Privacy and Security:
   - Adhere to strict privacy and security regulations to safeguard patient data.
   - Implement encryption, access controls, and other security measures to prevent unauthorized access and data breaches.

Requirement Specification

6. Integration with Clinical Workflow:
- The AI-based disease prediction system should seamlessly integrate with the clinical workflow, ensuring efficient utilization by healthcare professionals.
- It should offer timely and accurate predictions, enhancing clinical decision-making processes.

7. Continuous Improvement:
- The AI model should undergo continuous enhancement through ongoing monitoring and user feedback.
- Regular updates incorporating new data and algorithms are essential to improve the system's accuracy and performance over time.

In conclusion, the development of an effective AI-based disease prediction system for healthcare necessitates a comprehensive requirement analysis process. The system must be tailored to meet the specific needs and constraints of the healthcare industry while strictly adhering to privacy and security regulations. By adhering to these requirements, healthcare professionals can harness the power of AI to enhance disease prediction and ultimately improve patient outcomes.

Process Model: Iterative and Incremental Development

1. Define Objectives:
   - Identify and document the goals of our disease prediction system by engaging with stakeholders.

2. System Design:
   - Develop a comprehensive system design, considering scalability and flexibility.

3. Algorithm Selection:
   - Choose suitable machine learning algorithms (Decision Tree, Random Forest, Gradient Boosting) and fine-tune parameters.

4. Data Preparation:
   - Gather relevant datasets and preprocess data for training.

5. Model Development:
   - Implement algorithms, train models, and employ a Voting Classifier for ensemble predictions.

6. User Interface:
   - Design an intuitive and user-friendly interface for symptom input and result interpretation.

7. Testing:
   - Conduct rigorous testing for functionality, accuracy, and usability.

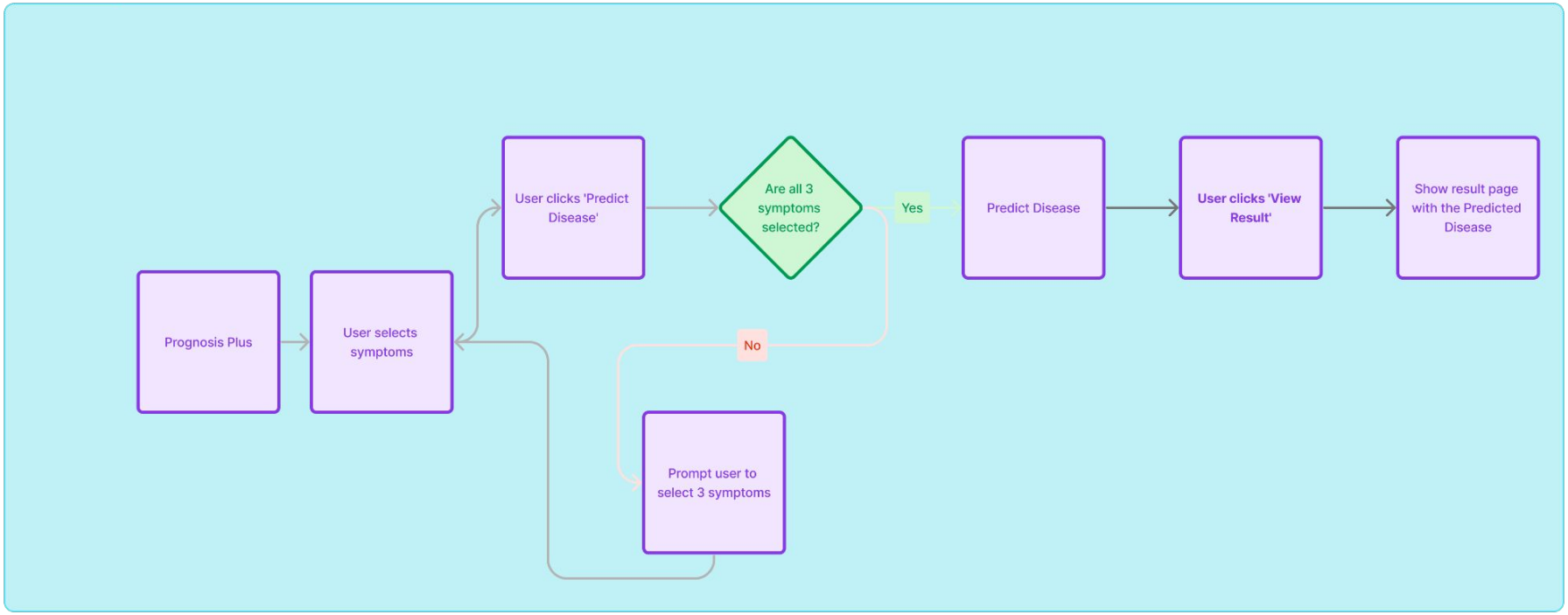
Process Model: Iterative and Incremental Development

8. Deployment:
   - Deploy the system to production and monitor performance.

9. Feedback and Iteration:
   - Collect user feedback and iterate on predictive models and the user interface.

10. Maintenance and Scaling:
    - Implement ongoing maintenance and scalability measures for continuous effectiveness.

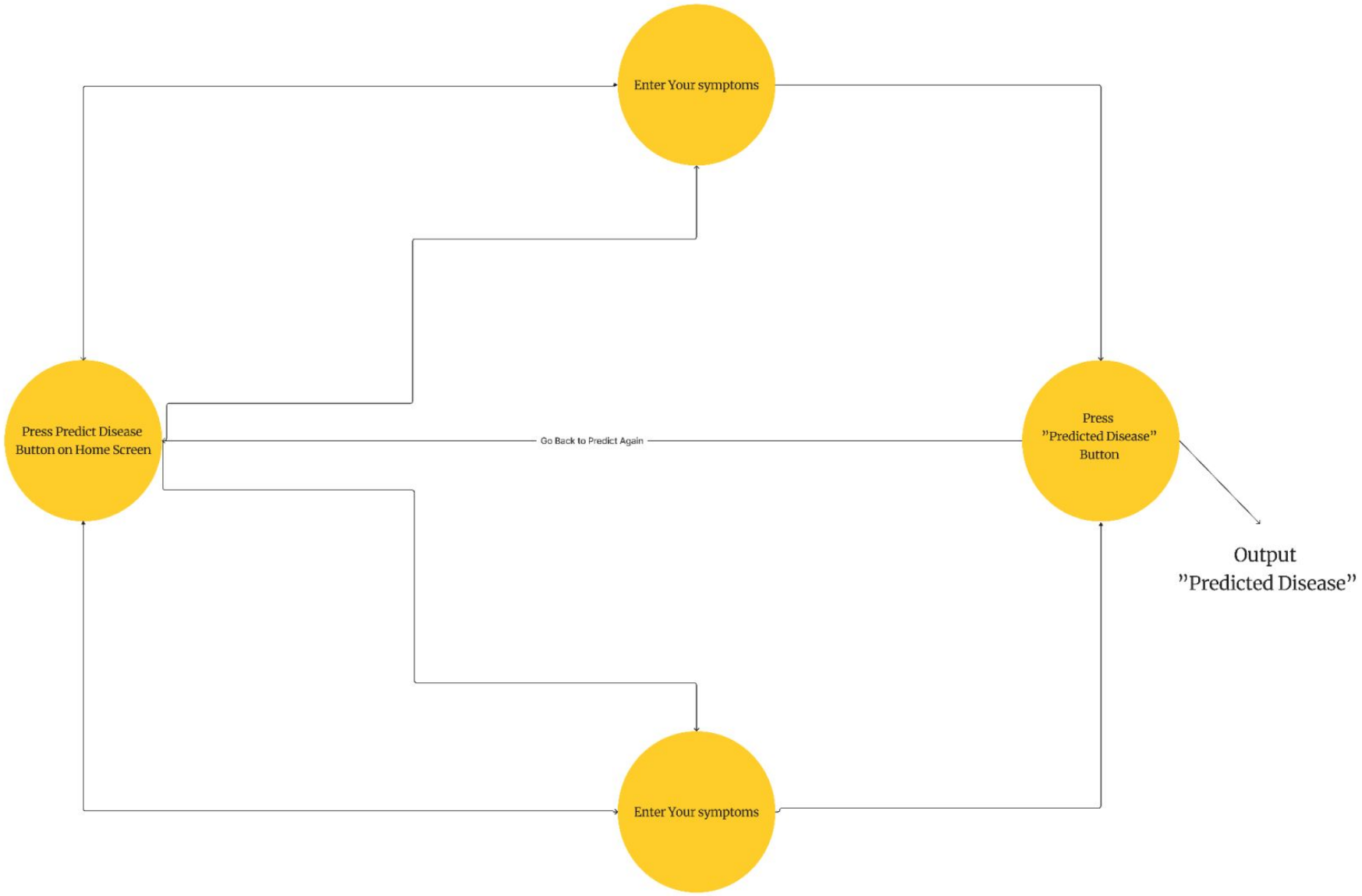
Block Diagram



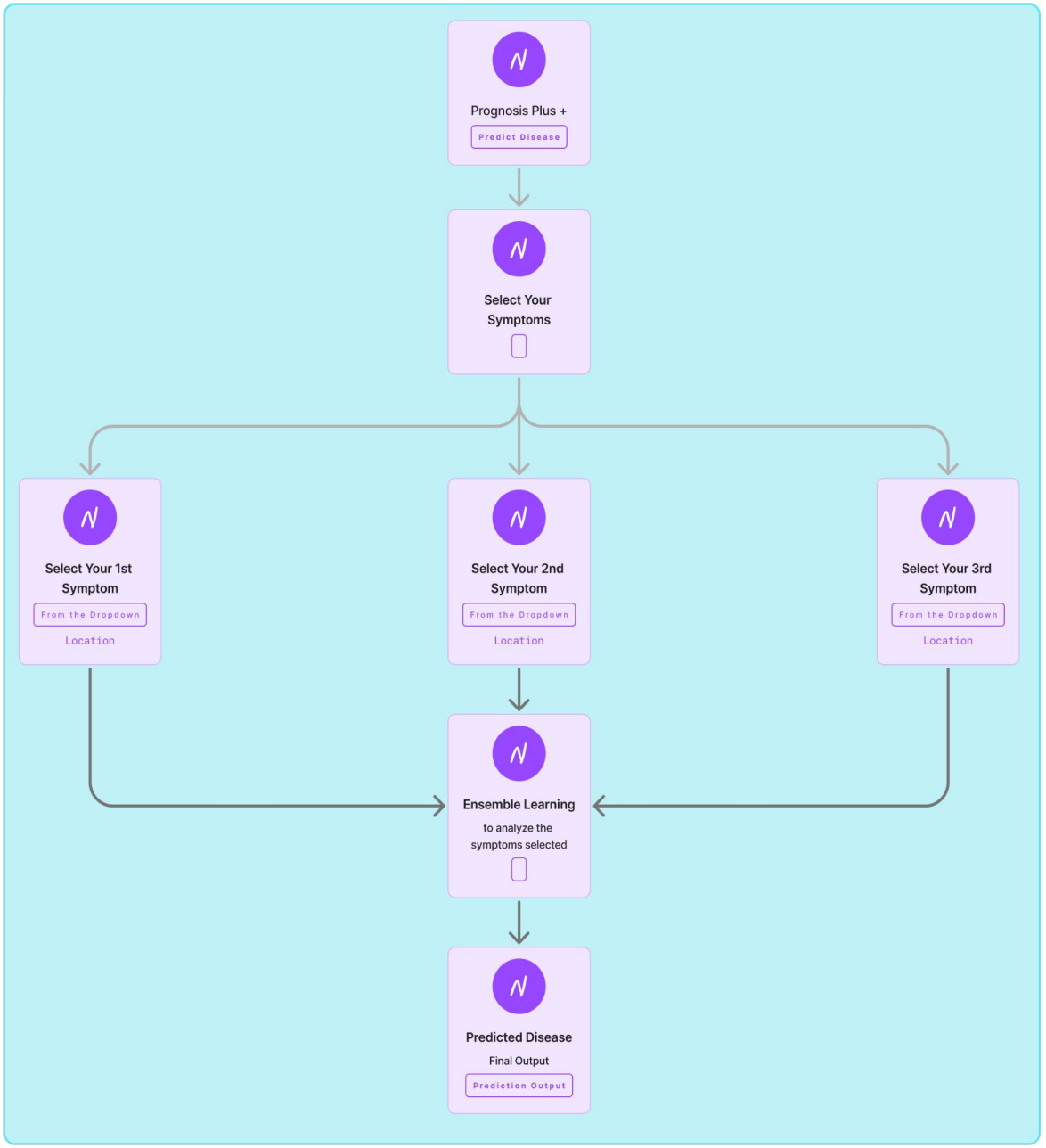
DESIGN SPECIFICATIONS

HTA, INTERACTION DESIGN(4 STEPS)STAKEHOLDERS IDENTIFICATION, STORYBOARDING,USE CASE MODELLING,PERSONAS,STATE TRANSITION NETWORK

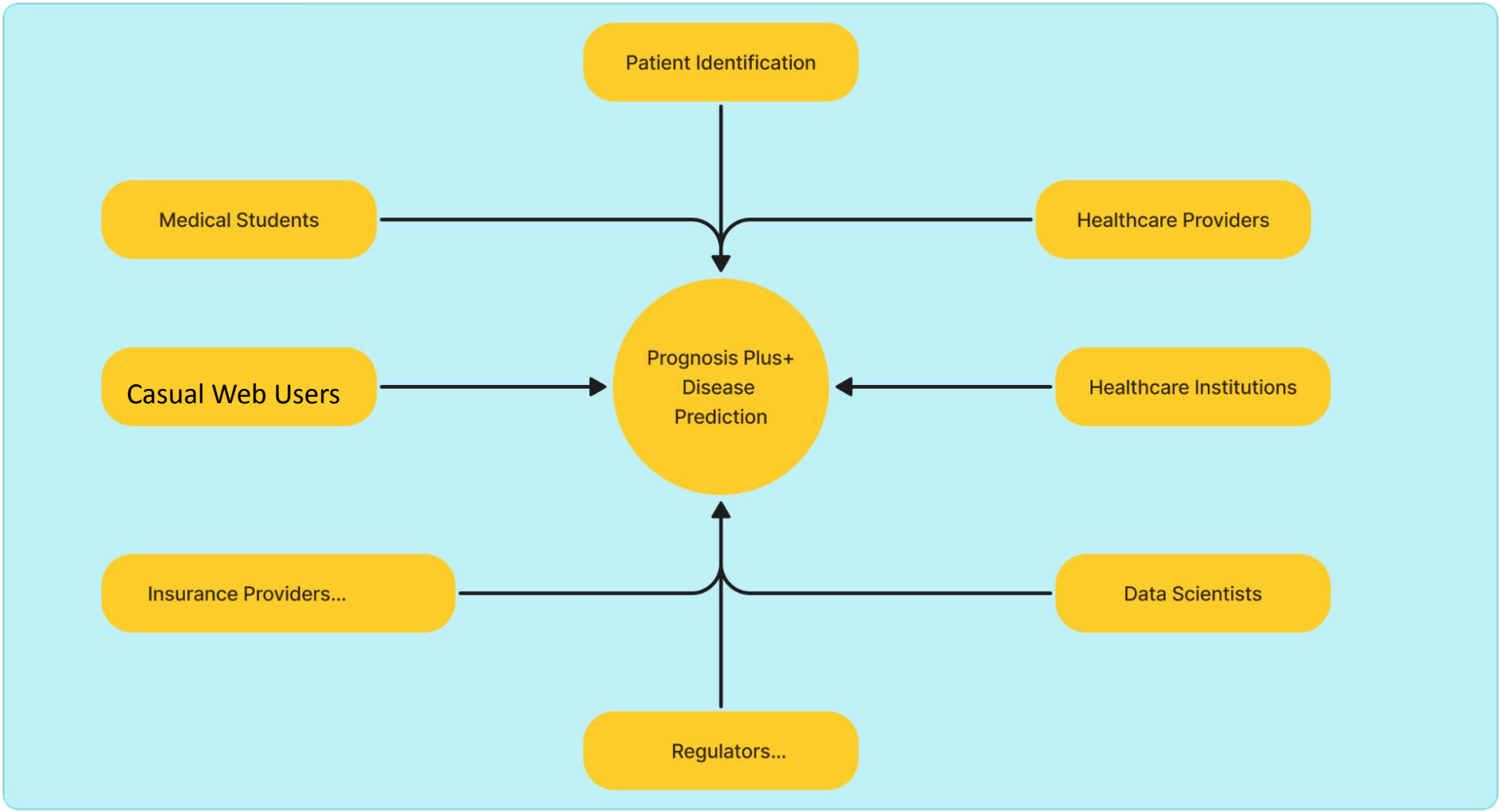
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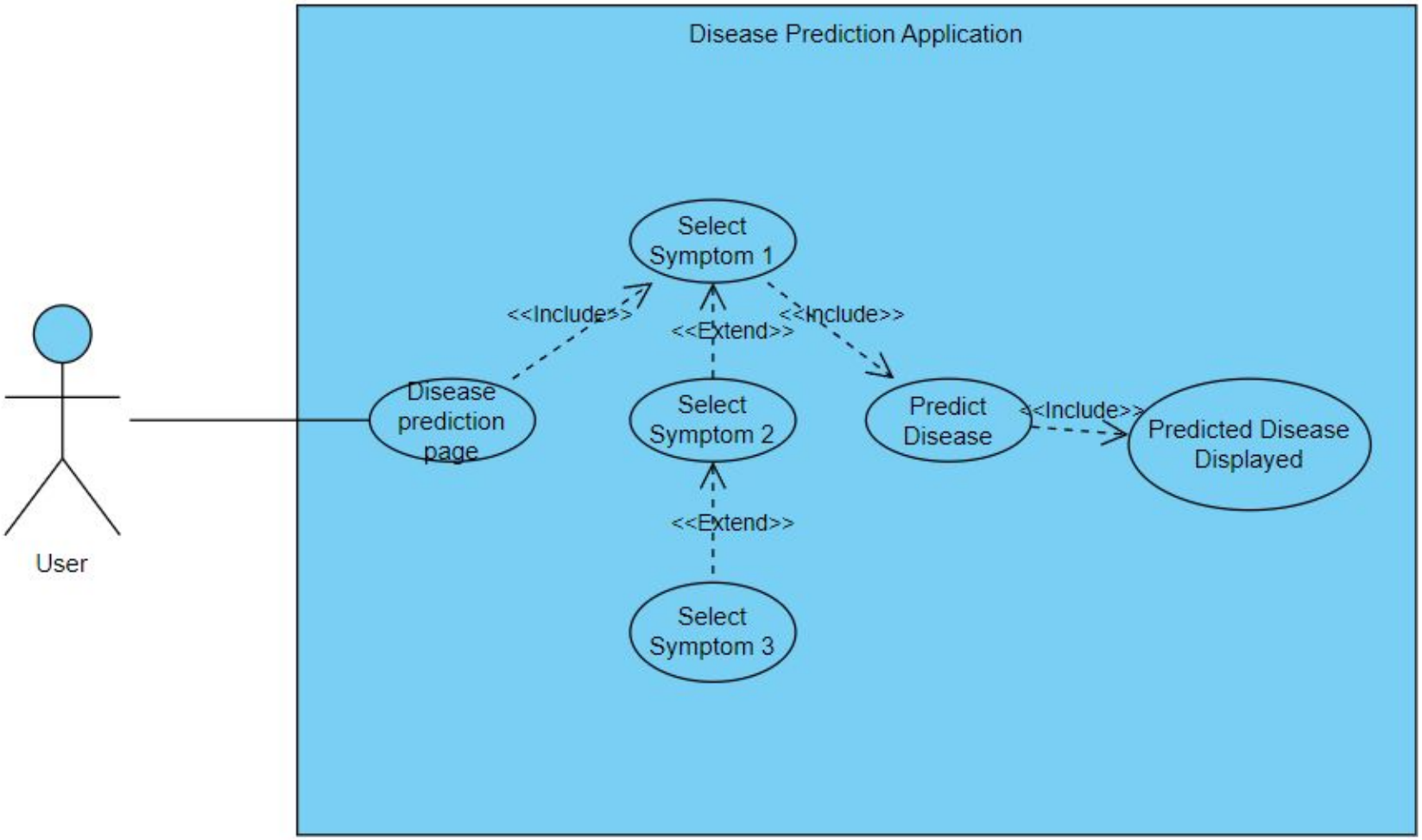
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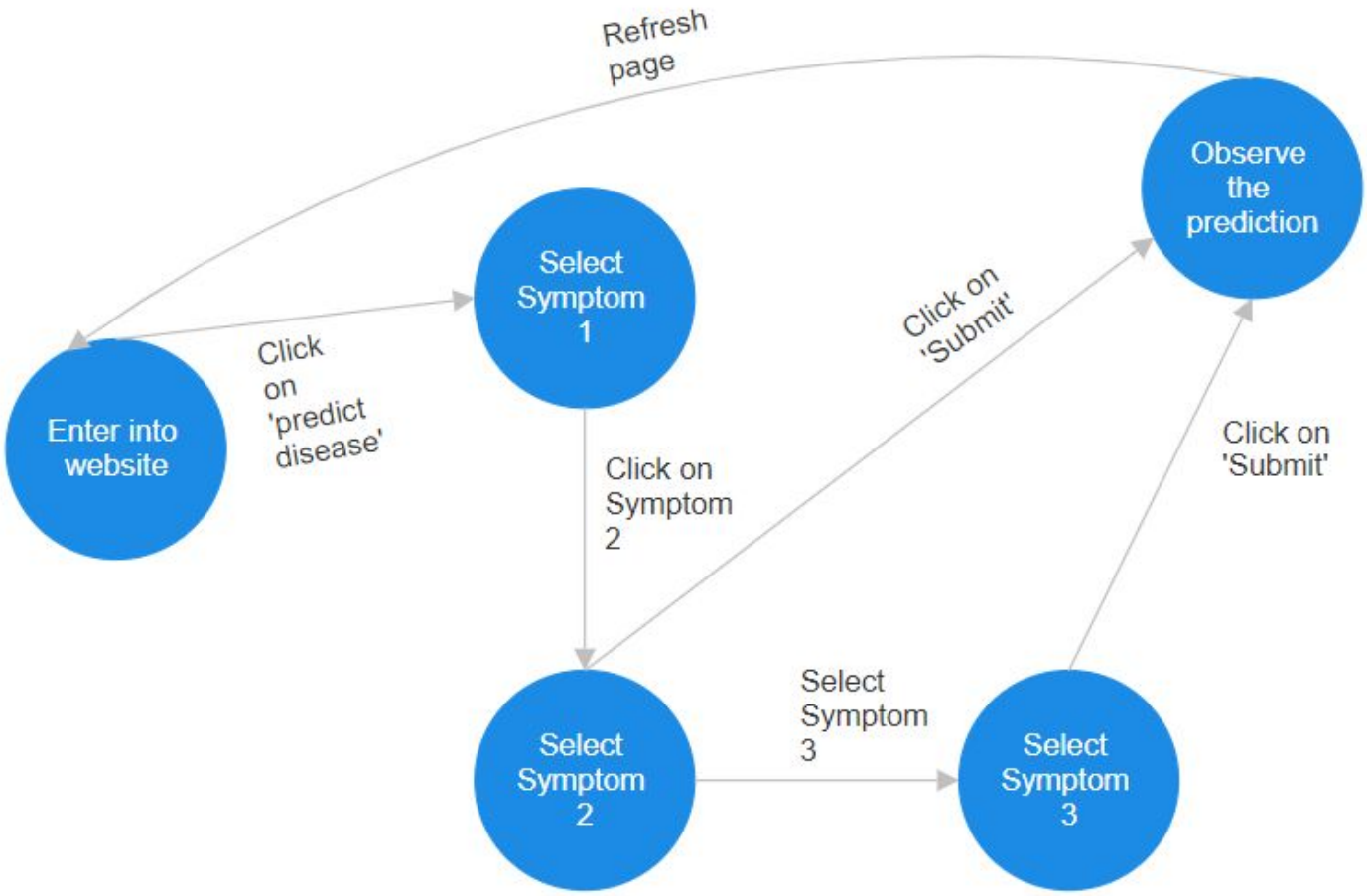
Stakeholder Identification



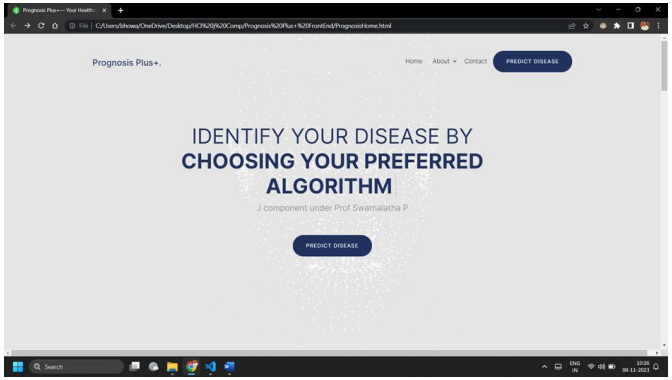
Use Case Diagram

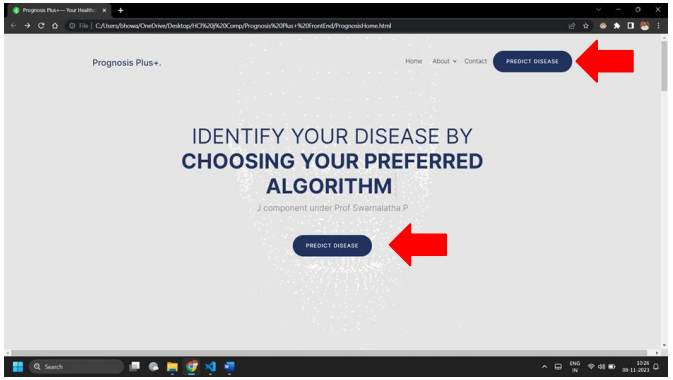


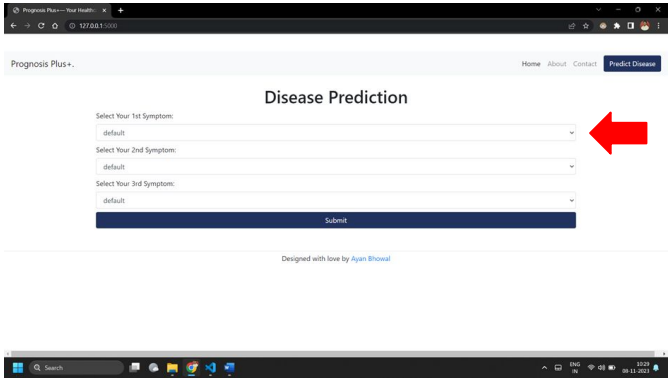
STATE TRANSITION NETWORK

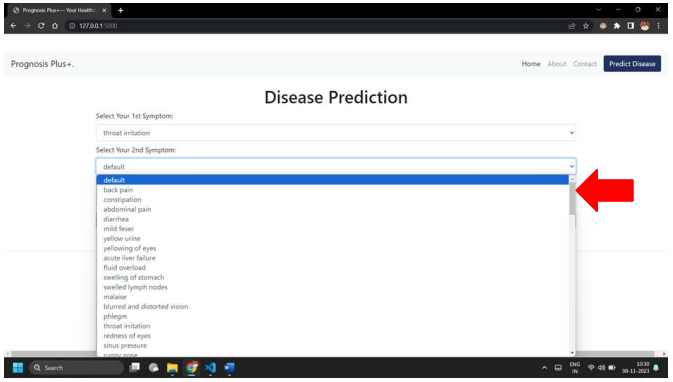


Story Board

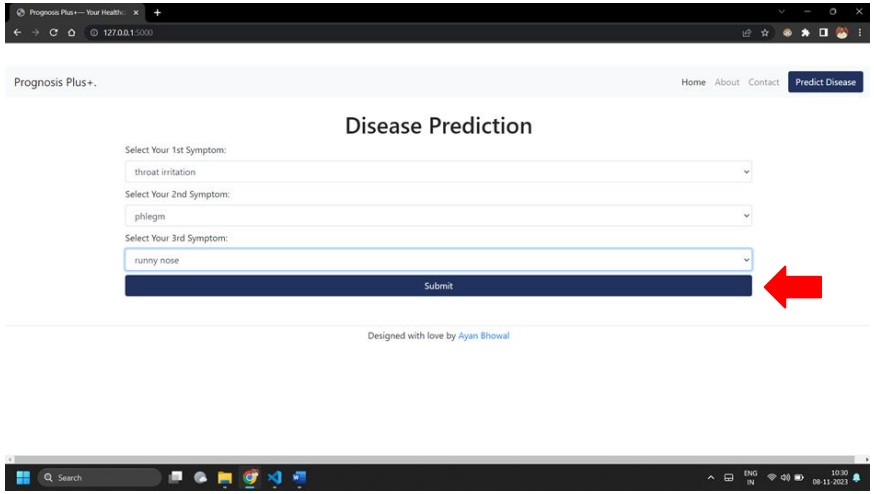

Prognosis Plus+.
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About
Contact
PREDICT DISEASE
IDENTIFY YOUR DISEASE BY
CHOOSING YOUR PREFERRED
ALGORITHM
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PREDICT DISEASE

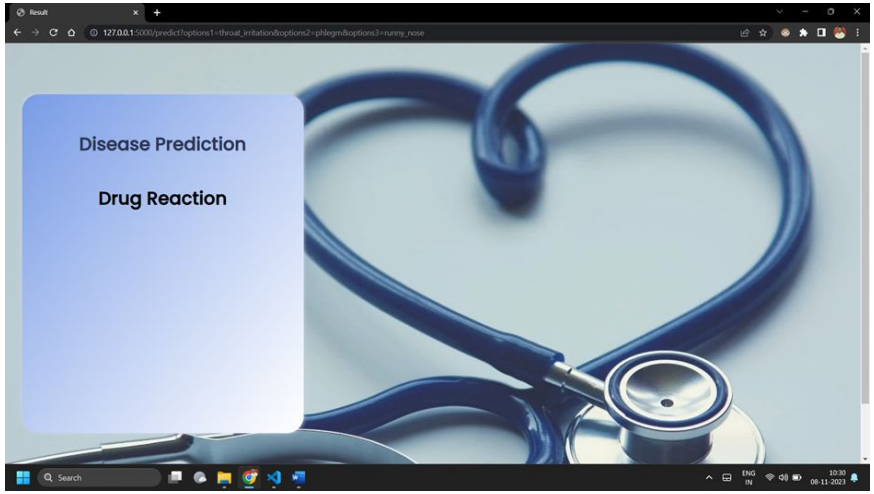

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default
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default
default
back pain
constipation
abdominal pain
diarrhea
mild fever
yellow urine
yellowing of eyes
acute liver failure
fluid overload
swelling of stomach
swelled lymph nodes
malaise
blurred and distorted vision
phlegm
throat irritation
redness of eyes
sinus pressure

Story Board


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Select Your 1st Symptom:
throat irritation
Select Your 2nd Symptom:
phlegm
Select Your 3rd Symptom:
runny nose
Submit
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Disease Prediction
Drug Reaction

Schneiderman's 8 Golden Rules

Strive for consistency	The utilization of icons and consistent color scheme within the website and its user interface significantly enhances the overall user experience (UX).
Enable frequent users to use shortcuts	Incorporated multiple call-to-action buttons to facilitate users in efficiently achieving their objectives within the user experience cycle.
Offer informative feedback	Upon completion, a subsequent page unfolds, furnishing the predicted disease as insightful feedback.
Design dialogs to yield closure	The conclusive page provides a clear and comprehensive description, affirming the successful prediction of the disease through precise and informative text. The landing page, adeptly designed, delivers a lucid explanation of the website's purpose and user-friendly usage guidelines.

Offer simple error handling	Users will receive prompts in case of errors while selecting symptoms from the dropdown menu, ensuring a smooth and error-free interaction.
Permit easy reversal of actions	Incorporated a "Go back" button to facilitate a seamless reversal of actions, enhancing user flexibility and navigation.
Support internal locus of control	The users can freely navigate through the website and their data will stay only visible to them. The predicted disease is only shown on the basis of the user’s choice of symptoms.
Reduce short-term memory load	The symptoms are chosen by the user from a list rather than having them recall and type it themselves.

GOMS Model

GOMS Model for Prognosis Plus+ Website Interaction:

1. Goal:
   - The primary goal is to predict a disease based on selected symptoms.

2. Operators:
   - P: Perception:
     - Recognize the need to visit the Prognosis Plus+ website.
     - Identify the homepage call-to-action buttons.

   - K: Knowledge:
     - Know how to operate a web browser.
     - Understand the purpose of the call-to-action buttons.

   - M: Motor:
     - Use the mouse to click buttons.
     - Navigate through dropdown menus.

   - S: Speech:
     - Use verbal feedback to express preferences.
     - No speech interactions are explicitly mentioned.

3. Tasks:
   1. Navigate to Prognosis Plus+ Website:
      - Subtasks:
        - Use the web browser (Perception, Knowledge).
        - Type the Prognosis Plus+ website URL in the address bar (Motor).

   2. Homepage Interaction:
      - Subtasks:
        - Observe and recognize the homepage call-to-action buttons (Perception).
        - Choose an action based on preference (Knowledge, Speech).
        - Click the selected button using the mouse (Motor).

   3. Redirect to Symptom Form:
      - Subtasks:
        - Understand the redirection process (Knowledge).
        - Follow the redirection to the form page (Motor).

   4. Symptom Selection:
      - Subtasks:
        - Observe and recognize three symptoms listed on the form (Perception).
        - Identify dropdown menus associated with each symptom (Perception).
        - Choose specific symptoms using the dropdown menus (Knowledge, Motor).

5. Predict Disease Button:
   - Subtasks:
     - Identify the "Predict Disease" button (Perception).
     - Click the button using the mouse (Motor).

6. Processing:
   - Subtasks:
     - Understand the processing phase (Knowledge).
     - Wait for the website to process selected symptoms (Perception).

7. Results Page:
   - Subtasks:
     - Observe the redirection to a page displaying the predicted disease (Perception).

8. View Prediction:
   - Subtasks:
     - Review the predicted disease based on selected symptoms (Perception).

9. Additional Information:
  - Subtasks:
    - Explore additional information about the predicted disease (Perception, Motor).

4. Selection Rules:
  - Selection rules involve choosing specific symptoms from dropdown menus and selecting actions based on personal preferences.

5. Execution Time:
  - The model does not provide specific information about the time required for each task.

6. Evaluation:
  - Evaluation involves recognizing visual elements, making decisions based on preferences, and reviewing predicted disease information.

7. Comments:
  - The model assumes a user with basic web browsing skills and does not include explicit speech interactions. Adjustments may be needed for users with different skill levels or accessibility requirements.

GOMS TABLE

GOMS Table

Task	Operators	Subtasks
1. Navigate to Website	P: Recognize, K: Know, M: Use, S: -	- Use the web browser (P, K). - Type the Prognosis Plus+ website URL in the address bar (M).
2. Homepage Interaction	P: Observe, K: Know, M: Click, S: Use verbal feedback	- Observe and recognize homepage call-to-action buttons (P). - Choose an action based on preference (K, S). - Click the selected button using the mouse (M).
3. Redirect to Form	P: Understand, K: Know, M: Follow	- Understand the redirection process (K). - Follow the redirection to the form page (M).

COMMUNICATION AND COLLABORATION

1. In the realm of collaboration and communication, our project enables multiple users to to identify their disease simultaneously.

2. Collaboration in HCI pertains to the ability of multiple users to work together on a task or project using technology. It involves systems or interfaces that facilitate teamwork, allowing individuals to coordinate, share resources, and contribute collectively towards a common goal.

3. The interface provided in our project is simple to understand by even a layman.

4. Groupware refers to software or technological tools designed to support and enhance group collaboration and communication.

5. These tools can include shared calendars, collaborative document editing platforms, video conferencing software, and other applications that facilitate communication and cooperation among individuals or groups.

USABILITY TESTING

1. Usability testing in HCI is a crucial process used to evaluate the ease of use and user experience of a digital product, software, or interface. It involves observing representative users as they interact with the system to identify usability issues, gather feedback, and make improvements.

2. We have tried to implement the same in our project by applying heuristics evaluation metrics, Schneiderman's 8 golden rules, questionnaires etc.

3. We requested our friends to give their feedback on the project.

4. We received some dissatisfaction regarding the user interface, prompting us to collaboratively adjust it based on their collective feedback. Ensuring adherence to the 8 golden rules guided our redesign of the current UI.

INTERFACE TESTING

1. Interface testing in human-computer interaction (HCI) refers to the evaluation and examination of the user interface of a software application, system, or device. This testing aims to assess various aspects of the interface's usability, functionality, and user experience. It involves scrutinizing how well the interface meets the needs of its intended users and ensures smooth interaction.

2. The interface holds significant importance in captivating customers' interest toward a specific product or service. Therefore, in our project, we ensured that the interface we developed aimed to grab the user's attention.

3. An approach to accomplish this is by crafting a visually appealing design. That's why we incorporated calming color schemes to enhance the interface's aesthetics.

4. Testing the interface's ability to handle errors or unexpected user inputs gracefully without crashing or causing confusion is also an important functionality.

5. Furthermore, the user receives their diagnosis almost instantly, ensuring a favorable response time for the user's convenience. This means that the system or service provides the user with their diagnosis or information extremely quickly, making sure that the time taken for the system to respond is very short, thereby enhancing the user experience. A quick response time is essential as it contributes to a smoother and more efficient interaction between the user and the system, preventing frustration or delays in obtaining important information or results.

USER ACCEPTANCE TESTING

1. User Acceptance Testing (UAT) is a critical phase in the software development lifecycle where the focus shifts to ensuring that the developed software or system meets the requirements and expectations of the end users. It's the final testing phase before the product is released to the users or customers. The primary goal of UAT is to validate the system's readiness for real-world use and to gain confidence that it satisfies user needs. Some of the key aspects of UAT are:-
2. End-User Involvement: UAT involves actual users or representatives of the end-users who test the software in a realistic environment or scenario.
3. Requirement Verification: UAT ensures that the software meets the specified requirements and functions as intended, addressing user needs effectively.
4. Real-World Scenarios: The testing process aims to simulate real-world scenarios, workflows, and use cases that users are likely to encounter.
5. Feedback Collection: Users provide feedback on usability, functionality, performance, and any deviations from their expectations.

CONT.

6. Validation of Business Processes: It verifies that the software aligns with the business processes and goals it was designed for.

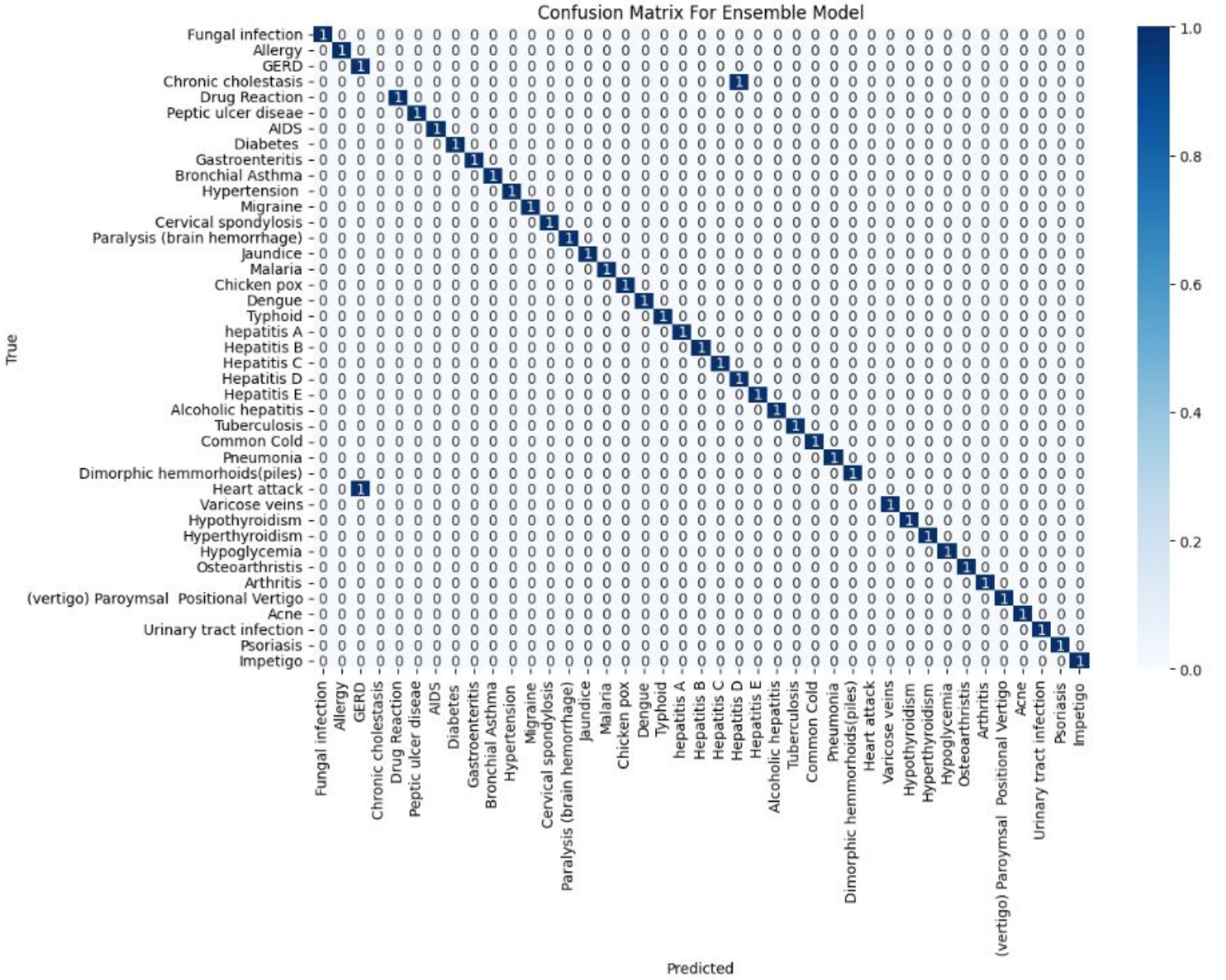
7. Bug Reporting and Resolution: Any issues, bugs, or discrepancies found during UAT are reported, tracked, and addressed before the final release.

Results

Accuracy	Precision	Recall	F1-Score
0.9512195121951219	0.9268292682926829	0.9512195121951219	0.9349593495934958

1. **Accuracy (95.1%):**
   Accuracy measures the overall correctness of our model.
   Our model correctly predicts the class for approximately 95.12% of the instances in the test set.
   High accuracy is generally good, but it might not be the only metric to consider, especially in imbalanced datasets.
2. **Precision (92.7%):**
   Precision measures our model's ability not to label as positive a sample that is negative.
   In our case, the model correctly identifies positive instances approximately 92.68% of the time.
   A high precision indicates that when the model predicts a positive class, it is likely correct.
3. **Recall (95.1%):**
   Recall (or sensitivity) measures our model's ability to capture all positive instances.
   Our model captures approximately 95.12% of the actual positive instances.
   A high recall indicates that the model is effective in identifying most of the positive cases.
4. **F1 Score (93.5%):**
   The F1 score is the harmonic mean of precision and recall. It balances false positives and false negatives.
   Our F1 score of approximately 93.50% suggests a good balance between precision and recall.
   This is particularly useful when we want to avoid either too many false positives or too many false negatives.

Confusion Matrix



Confusion Matrix

The presence of a diagonal of 1s in our confusion matrix is a positive indication for our model's performance. Each element along the diagonal represents the number of instances that were correctly classified for each class, or in other words, the true positives (TP).
This pattern suggests that our model is making accurate predictions across various classes, demonstrating its ability to correctly identify and distinguish between different diseases or conditions. The higher the values along this diagonal, the more instances our model has successfully classified.
In summary, a strong diagonal presence in the confusion matrix signifies that our model is performing well in terms of correctly predicting the classes across the different categories, showcasing a robust and accurate classification capability.

CONCLUSION

In conclusion, our disease prediction system, incorporating ensemble learning with Decision Tree, Random Forest, and Gradient Boosting algorithms, represents a significant advancement in healthcare technology. Through an iterative and user-centric development process, we have successfully created a predictive model that not only achieves high accuracy in disease prediction but also offers a seamless and intuitive user interface.

Performance Improvement:
Through rigorous testing and iterative refinement, we have enhanced the performance of our system. Fine-tuning algorithm parameters, optimizing data preprocessing, and constantly revising the usability has contributed to the system's accuracy and reliability. The ensemble approach, combining the strengths of multiple algorithms, has proven to be a robust solution, providing a more comprehensive and accurate disease prediction capability.

Real-Time Applicability:
One of the key strengths of our system lies in its applicability in real-time environments. The efficiency of the algorithms, coupled with the user-friendly interface, allows for quick and reliable predictions. In a real-time setting, our system can aid healthcare professionals and individuals alike in making prompt decisions regarding disease prevention and management.

CONCLUSION

Future Directions for Improvement:
To further elevate the system's performance, future enhancements could explore additional algorithms, continuous learning mechanisms, and incorporation of emerging medical data. Machine learning models, by nature, can benefit from ongoing updates and adaptations to stay current with evolving medical knowledge and symptom patterns. Hence, testing the data with stronger models, unsupervised learning, neural networks etc, can raise the efficiency and ease of usability.


Thank
You